

Long Run Market Access and the Rise of the Sunbelt*

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Abstract

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*People to thank...

1. Introduction

- Introduction
- Historical Background
- Data and Summary Statistics
- Empirical Analysis and Results
 - Urban wage premium, number of workers, and market access
 - * relative to housing values
 - Urban skill premium, number of workers, and market access

2. Data

To estimate the relationship between the urban wage premium and the role that market access has on the premium throughout the post war era, we draw on several datasets. First, to measure market access, we construct a new harmonized decadal dataset of the domestic US Highway Network that we describe in more detail below. To measure the average earnings per worker (wages), we use the IPUMS Full Count Census for 1940, the 1 percent sample Census for 1950 and 1970, the 5 percent sample Census for 1960, 1980, 1990, and 2000, and the 2005, 2010, and 2015 American Consumer Surveys. Our sample consists of adult men living in the continental US between the ages of 25 and 65 who are employed for at least 26 weeks in the sample year. We apply IPUMS person weights in our summary statistics and all our regressions.

We compute average weekly earnings (wages) by finding total income divided by the number of weeks worked in the given survey year. For individuals earning the top income code, we set their total income to the lower top

income cutoff multiplied by 1.5.¹ If the urban wage premium is primarily driven by people in the top income group, our estimates of the urban wage premium will be biased downwards by this choice. All incomes are then multiplied by the Consumer Price Index to be in 2015 dollars. In order to have a consistent measure of occupation, we use the ‘occ1950’ variable provided by IPUMS which harmonizes occupation codes across all our survey years.

2.1 Market Access Data

To construct a measure of market access, we follow the inter-regional trade literature developed in in Eaton and Kortum (2002), Redding and Venables (2004), and Donaldson and Hornbeck (2016).² A county c ’s market access, MA_c , is given by this system of non-linear equations

$$MA_c = \rho \sum_d \tau_{cd}^{-\theta} MA_d^{-1} Y_d, \quad (1)$$

where θ is the trade elasticity with respect to trade costs, ρ is a constant from the theoretical model, and Y_d is the income of county d measured by GDP.

Our measure of the iceberg trade costs to ship from county c to county d in year t , τ_{cdt} , measures the fuel costs to travel to a county d as well as the wage paid to the truck driver for the drive time. The formula for τ_{cdt} is given by

$$\begin{aligned} \tau_{cdt} = & \text{distance in miles}_{cdt} \times \text{cost per mile}_t \\ & + \text{travel time in hours}_{cdt} \times \text{hourly wage of truck drivers}_t \end{aligned} \quad (2)$$

To solve the system of non-linear equations derived from the model re-

¹These cutoffs are reported in IPUMS INCWAGE documentation. In 2015 \$, the cutoffs are \$84,500 for 1940, \$99,200 for 1950, \$199,500 for 1960, \$309,000 for 1970, and \$225,000 for 1980. Afterwards, IPUMS reports income as the median wage above that year’s cutoff.

²The particular trade model used is described in Jaworski and Kitchens (2019).

quires detailed trade cost, τ_{cd} , from every origin to each destination. We approximate the origin-destination trade costs in each year by measuring travel costs through the contemporary highway network. Our measure of trade costs relies on measuring the distance and time traveled through the highway network, which is then monetized using hourly wages and fuel cost in each period, as highlighted above. To measure the time and distance post 2000 (2005 and 2015), we rely on the detailed shapefiles maintained by the federal government. For years prior to 2000 we georeference atlases produced by Rand McNally in each decade 1940-1990, as the federal government did not begin maintaining its database until 1994. To create a time consistent network, we begin with the 2000 version of the Federal Highway Administration’s shapefiles. Then, for each decade, we overlay scans of each Rand McNally atlas and edit the federal shapefile. Specifically, we delete road segments that were constructed between the current map year and 2000, and re-code segments that have been reclassified over time (i.e., segments of the US Highway network that became interstates, or paved roads that were previously gravel or dirt). We perform this procedure iteratively for each decade back to 1940. One advantage of this approach is that it locates roads based on modern GPS technologies to ensure accurate placement throughout time.³ We then follow the process outlined in Jaworski and Kitchens (2019) to measure the point to point distance travel time through the network each decade. That is, we represent each location by its geographic centroid, connected to the network by a set of access roads. We assign a travel speed to each segment in the network according to its classification (state, federal, interstate) and surface (dirt, gravel, paved), according to the parameters in 1. We then solve for the least cost route through the network for all pairwise combinations of origins and destinations.⁴

The market access variable summarizes a county c ’s access to the trans-

³One drawback is that this approach may ignore roads that were important prior to the

Table 1: Road Classifications and Speed Parameters

Class of Road	Speed limit	Speed limit
Interstate	65	70
US Highway (Paved)	45	55
US Highway (Gravel)	35	50
US Highway (Dirt)	30	45
State Highway (Paved)	25	45
State Highway (Gravel)	20	40
State Highway (Dirt)	15	35
Access Road	10	10

Notes:

portation network by taking an average of other counties' market size weighted by both the trade cost to ship to that market and by all other counties access to that market (MA_d^{-1}). Therefore a county c 's market access is large when it has access to large markets that other counties have little access to.

To aggregate market access to Urban areas, we created a crosswalk of counties within MSAs. For each MSA, we average across the counties within the MSA.⁵ Our measure of non-urban market access for a given state is the average of counties in that state that do not fall within any MSA. Therefore, for each individual we have a measure of whether or not they live in an urban area and the market access of the urban/non-urban area they live in. This process is detailed in Appendix B.

modern stock of infrastructure.

⁴More details on how the trade costs are calculated can be found in Section III of Jaworski and Kitchens (2019).

⁵Our results are robust to weighting by population. (NEED TO CHECK)

Table 2: Summary Statistics

Year	N	% Urban	log(Market Access)		Urban Weekly Wage (2015 \$)		Nonurban Weekly Wage (2015 \$)	
			Mean	SD	Mean	SD	Mean	SD
1940	16,939,585	0.66	8.20	1.59	33	20	25	17
1950	68,457	0.69	9.33	1.29	72	41	59	33
1960	1,364,970	0.66	9.92	1.24	124	77	100	60
1970	280,338	0.75	10.49	1.16	205	138	166	105
1980	1,910,440	0.73	10.70	1.04	409	284	341	221
1990	2,262,149	0.74	12.06	0.81	701	562	558	405
2000	2,583,088	0.79	12.35	0.74	1,008	1,015	767	677
2005	530,737	0.79	12.33	0.75	1,143	1,146	863	712
2010	530,783	0.80	12.29	0.75	1,228	1,210	961	819
2015	561,934	0.81	12.28	0.77	1,353	1,446	1,068	1,011

Notes: Summary statistics weights individuals by their person weight. Weekly wage is reported in 2015 \$. Market access is calculated from solving $MA_c = \rho \sum_d \tau_{cd}^{-\theta} MA_d^{-1} Y_d$, where transportation costs, τ_{cd} , are calculated as described by Equation 2.

2.2 Summary Statistics

In Table 2, we report summary statistics of key variables. The first thing to note is that in the post war era, the share of people living in urban areas has steadily increased. Between 1940 and 2015, there has been a 15 percentage point increase in the urban share, from 66 percent in 1940 to 81 percent in 2015. This trend matches those found in Boustan et al. (2013). The log of market access also increases in the post war era, until about the year 2000, when it levels off. Market access increases in part due to major improvements in the transportation network, such as the expansion of the Interstate Highway System (IHS), Appalachian Development Highway System (ADHS), and other improvements in the state and federal aid highways that increase driving speeds. Aside from improvements in network infrastruc-

ture, the growth in market access reflects GDP growth, which has grown from 1.3trillionin1940to18.2 trillion in 2015. The last four columns of Table 2 show the wage among people living in MSAs versus those that work in nonurban areas. Both urban and nonurban mean weekly wages are growing over time in 2015 \$ from \$33.24 and \$25.23 in 1940 to \$1353.95 and \$1067.48 in 2015.

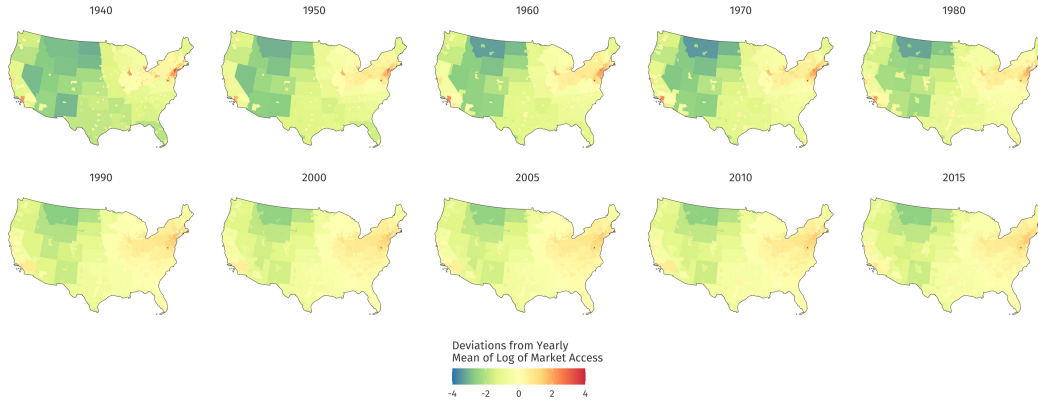
Figure 1 shows within-year deviations of the log of market access from the yearly mean at the MSA-level and the state average of non-urban counties. The figure shows that over time, non-urban areas have had a relative growth in market access, particularly in the southern and western United States. On the other hand, large cities in the 1940 had very large market access relative to the national average and this relative advantage declines over time from about 4 log points to about 2 log points above the mean. The deviation in log market access from the national average highlights the variation used in our regressions of the log of wage on the log of market access.

While the southern and western states have experienced relative gains in market access compared to the northeast, market access has increased everywhere throughout the 2nd half of the 20th Century. In Figure 2 we document how the log of market access has changed over time. For example, North Dakota in 2015 has about the same level of market access that New York City had in 1940.

@Question: For the map, there are some msas that are in the list we have but not showing up in the ipums data, should I use them still in the map?

@Question: Should there be a robustness check to prevent selection over time. If smaller and smaller MSAs are added, then we would expect a decline in wage premium even if the original cities' premium perhaps is growing. Also, it could be that our market access variable is picking up, in part, differences in the urban wage premium based on city size.

Figure 1: Log of Market Access Deviations from Yearly Mean: 1940-2015



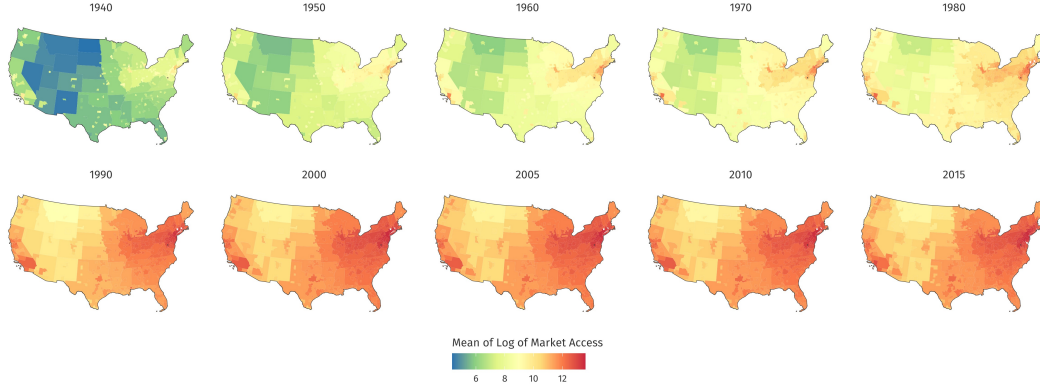
Notes: This figure shows deviations from the national mean of log market access averaged to MSAs and the state-wide average of non-urban counties. These averages are described in Appendix B. Market access is calculated from Equation 1 using trade costs calculated in Equation 2.

3. Results

For an initial understanding of the urban wage premium, for each survey year we take the ratio between the average urban weekly wage and the nonurban weekly wage. This ratio replicates the method used in Boustan et al. (2013) to calculate the urban wage premium. In Figure 3, the grey line with square markers shows the results of this exercise. The U-shape in which the urban wage premium starts around 30% in 1940, decreases between 1950 and 1980 to just over 20% and then increases again to over 25% after 2000 is consistent with results found in the literature.⁶

⁶Our results match the shape over time found in Figure 3 of Boustan et al. (2013). Our estimate in 2000 is a few percentage points lower than the estimate of 32% from Glaeser and Maré (2001). The increase between 1980 and 2000 matches Baum-Snow and Pavan (2012).

Figure 2: Log of Market Access over Time: 1940-2015



Notes: This figure shows the log market access averaged to MSAs and the state-wide average of non-urban counties. These averages are described in Appendix B. Market access is calculated from Equation 1 using trade costs calculated in Equation 2.

Moving into a regression framework, we run versions of the following specification separately for each survey year:

$$\log(\text{Weekly Wage}_i) = \beta_0 + \beta_1 \text{Urban}_i + \beta_2 \log(\text{Market Access}_{MSA(i)}) + \delta X_i + \varepsilon_i, \quad (3)$$

where X_i is a set of controls consisting of age in 5-year bins, education levels, and an indicator for being White. Our estimate of β_1 is the percent increase in wages that workers in urban areas have on average which we call the urban wage premium. The first regression only includes an indicator for living in an urban area without the log of market access or controls X_i . This is the regression based version of the Boustan et al. (2013) approach. However, there can be demographic differences between urban and rural areas that contribute to wage differentials and so we add controls for education, race,

and age in the following specifications.

The wages of individuals in Urban areas also benefit from the improved transportation network, so in the final specification, we add in the log of market access. After controlling for market access, the coefficient on Urban, β_1 should pick up on the non-transportation causes of the urban wage premium.⁷

Our estimates for the urban wage premium will be biased if there is sorting into cities by characteristics that affect wages but are distinct from age, race, and education. An example could be that workers sort into cities based on the industry they work in and differences in industrial wages can explain some of the remaining wage premium. Combes et al. (2008) show using panel data that individual fixed effects remove a sizeable portion of the variation in urban wage premium. Our time-series data is cross-sectional, so we can not use individual level fixed effects in our estimation. For our estimates to be causal, the exclusion restriction requires there to be no sorting beyond age, race, and education levels.

The results from the regressions that do not include a measure of market access is found in Figure 3 as the black lines with a diamond and a triangle marker. All regression tables can be found in Table A.1. These lines show that the urban wage premium still decreases from a premium of around 30% in 1940 through 1980. However, instead of the increase in the modern economy, our regression-based results show that the urban wage premium roughly levels off after 1980 at 20% without controls and at around 16% when controlling for age, education, and race. This result suggests that increased sorting of high-earning individuals into cities can explain the recent increase in urban wage premium found in the raw-measures.

The final specification includes market access in the regression of the log

⁷This includes the improved human capital development from Glaeser and Maré (2001) and improved firm-worker matching found in Baum-Snow and Pavan (2012).

of weekly wage on an urban indicator variable. This specification controls for the benefits cities receive from being located in an advantageous location in the transportation network. If part of the wage premium is explained by better location in the transportation network, we expect the premium to decrease when including market access which summarizes the network. The estimates on the urban indicator are found in Figure 3 as the black line with circle markers and in Table 3.

When controlling for the transportation advantage of cities and demographic differences, we see the urban indicator in 1940 has decreased to about 15% from the 30% observed in specifications that don't include market access. There still exists, albeit a much smaller, decline in urban wage premium through the 80s which flattens out to an urban wage premium that floats around 10% until at least 2015.

This result suggests that between 25.9% to 47.7% of the urban wage premium is explained by a relatively strong position in the transportation network.⁸ In particular, it appears that early in our sample, the differences in access to transportation network was large between urban and nonurban areas. This matches the fact that historical highway networks focused on connecting large cities and starting around 1955 began connecting many more places which increased market access for less populous areas (as evidenced by Figure 1).

⁸The % change in the urban wage indicator between the specification including controls and the specification including controls and market access is 47.7% for 1940, 30.7% for 1950, 35.9% for 1960, 46.2% for 1970, 35.4% for 1980, 37.4% for 1990, 26.1% for 2000, 25.8% for 2005, 33.1% for 2010, and 27.4% for 2015.

Table 3: Urban Wage Premium Estimates by Survey Year

	1940	1950	1960	1970	1980	1990	2000	2005	2010	2015
Urban	0.147 (0.010)	0.137 (0.012)	0.130 (0.011)	0.098 (0.011)	0.086 (0.011)	0.100 (0.011)	0.116 (0.011)	0.112 (0.011)	0.085 (0.014)	0.076 (0.017)
$\log(MA)$	0.053 (0.003)	0.035 (0.007)	0.051 (0.010)	0.056 (0.009)	0.038 (0.010)	0.093 (0.011)	0.070 (0.010)	0.068 (0.013)	0.067 (0.015)	0.054 (0.016)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934

Notes: Each cell is the coefficient estimate corresponding to the regression for the survey year indicated by the column label. The dependent variable is log weekly earning and each regression includes an indicator for living in an urban area, our measure of log market access, and controls for age in 5-year bins, education levels, and an indicator for being White. All regressions are weighted by the IPUMS person weights. Standard errors are in parenthesis and are clustered at the MSA-Nonurban State level

(@QUESTION: Not sure how to word the standard errors. They cluster at the MSA level and also at the nonurban state level.)

References

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Table A.1: All Urban Wage Premium Estimates by Survey Year

	1940	1950	1960	1970	1980	1990	2000	2005	2010	2015
Panel A: Urban only										
Urban	0.3153 (0.0177)	0.2208 (0.0131)	0.2295 (0.0135)	0.2109 (0.0119)	0.1692 (0.0110)	0.1943 (0.0157)	0.2018 (0.0144)	0.2014 (0.0160)	0.1759 (0.0165)	0.1661 (0.0175)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934
Panel B: Urban with Controls										
Urban	0.3153 (0.0177)	0.2208 (0.0131)	0.2295 (0.0135)	0.2109 (0.0119)	0.1692 (0.0110)	0.1943 (0.0157)	0.2018 (0.0144)	0.2014 (0.0160)	0.1759 (0.0165)	0.1661 (0.0175)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934
Panel C: Urban, Controls, and Market Access										
Urban	0.147 (0.010)	0.137 (0.012)	0.130 (0.011)	0.098 (0.011)	0.086 (0.011)	0.100 (0.011)	0.116 (0.011)	0.112 (0.011)	0.085 (0.014)	0.076 (0.017)
$\log(MA)$	0.053 (0.003)	0.035 (0.007)	0.051 (0.010)	0.056 (0.009)	0.038 (0.010)	0.093 (0.011)	0.070 (0.010)	0.068 (0.013)	0.067 (0.015)	0.054 (0.016)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934

Notes: Each cell is the coefficient estimate corresponding to the regression for the survey year indicated by the column label. The dependent variable is log weekly earning and each regression includes an indicator for living in an urban area. Panel A just includes the Urban indicator; Panel B includes controls for age in 5-year bins, education levels, and an indicator for being White; Panel C includes the above controls and our measure of market access. All regressions are weighted by the IPUMS person weights. Standard errors are in parenthesis and are clustered at the MSA-Nonurban State level

A. Full Urban Wage Premium Regressions

B. MSA-level Market Access

We aggregate our county-level measure of Market Access from Equation 1 to a measure at the urban area. For each year, we start with a crosswalk provided by IPUMS that lists which counties fall (at least partially) inside the boundaries of the metro area. Our metro area boundaries are the 1950 Standard Metropolitan Area for 1940 and 1950, the Metropolitan Statistical Areas for their respective year in 1960-2010, and the Core-Based Statistical Area for 2015.

This list is modified slightly to include MSAs that are recorded in the IPUMS survey data but are not included in that year's crosswalk. The counties that the MSA covers are identified by hand using the shape files from NHGIS for that year or if missing from the previous year.

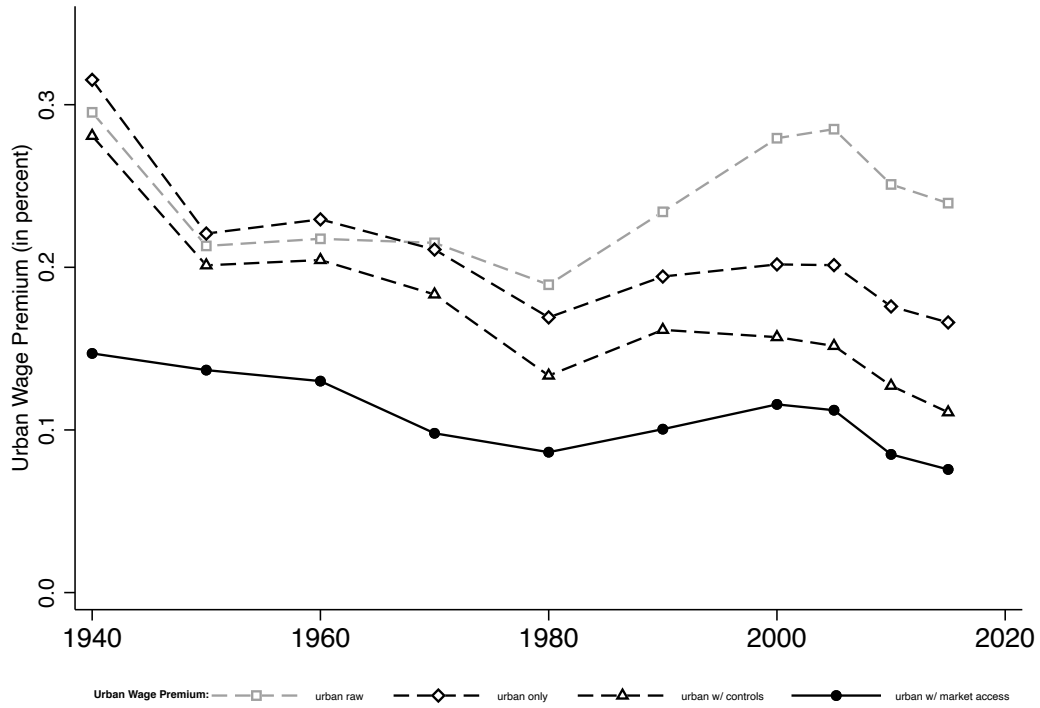
For a given year, a metro area's market access is the geometric average of the corresponding counties' market access. Then, for our measure of market access for the nonurban portion of a state is the geometric average of all counties that are not within any metro areas. We include robustness checks for weighting the average by county population.

B.1 Robustness Checks

B.1.1 MA Weighted by Population

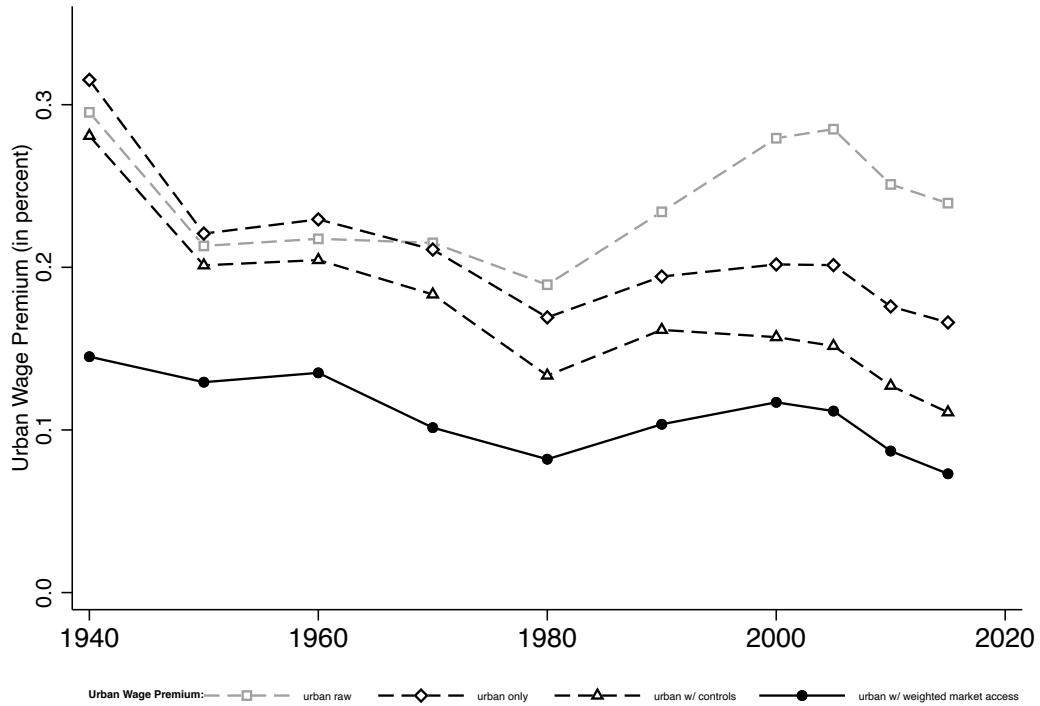
For some metro areas, there is a central county and more counties at the peripheral, less populated counties. In this case, our simple average may understate the market access in a city, so we check for this by averaging across counties weighted by the counties population share. The results in Figure 1 show that the shape of the black line with circle markers is very similar to Figure 3.

Figure 3: Urban Wage Premium 1940-2015



Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 3. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of market access.

Figure 1: Urban Wage Premium 1940-2015 with Weighted Market Access



Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 3. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access.